

DANIEL DUCLOS-CAVALCANTI

Systems Software Engineer

New York, New York

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Experience

Software Engineer II

Mar 2025 – Present

Ripple

New York, USA

Managed Custody

Mar 2026 – Present

- One of **3** engineers building a complete **greenfield** managed-custody product: a **Rust** orchestration layer over Ripple Custody on **AWS Lambda**, **Step Functions**, **EventBridge**, **DynamoDB**, and **KMS** for tier-1/2 bank clients.
- Owned the **actors**, **approvals**, **events**, and **domains** APIs: **casbin**-gated maker/checker **quorums**, per-actor **KMS** signing keys bound by **proof-of-possession**, and an **event-sourced**, cursor-paginated stream.
- Built and managed scalable infrastructure using **AWS CDK**: **35 stacks**, ~1,100 resources per environment, branch-scoped naming allowing highly concurrent **ephemeral environment** deployments with auto-tear-down.
- Architected an **AI-native software factory**: a crew of role-specialized agents (**plan** → **implement** → **review** → **merge**) on a cron-driven state machine semi-autonomously shipping **Jira** tickets to production, lifting delivery ~10× (**15**→**155** **MRs/mo**, peak **205**) across the product's ~**6-month** build.

Ripple Custody (Core Platform)

Mar 2025 – Mar 2026

- Migrated a **HSM**-backed key **cold-storage** custody service (vault) from legacy **C#/.NET** to a **Rust** (**axum**) drop-in replacement, serving its batch file-transfer execution API over **gRPC** to two live customers.
- Developed a benchmarking harness in **Rust** (**criterion**) to profile cryptographic operations across the custody's **KMS** backends (**IBM** and **Luna HSMs**, **MPC**), tracking performance regressions in **CI**.

Research Assistant

Jul 2022 – Oct 2022

TU Munich

Munich, Germany

- Profiled ML inference across CPUs, GPUs, and Coral Edge TPUs (TensorFlow Lite) via USB traffic analysis (PyShark).
- Established up to 57% of inference time is data transmission to the TPU, informing TensorDSE's hardware mapping.

Embedded Software Engineer – Part Time

Aug 2021 – Jan 2022

Molabo GmbH

Ottobrunn, Germany

- Built simulation tooling via Linux virtual CAN, raising GTest coverage 25% on safety-critical motor-controller firmware.
- Owned the team's Jenkins/Docker/CMake **CI** pipeline, supporting 10+ engineers.

Tutor (Embedded Systems Programming Lab)

Apr 2021 – Aug 2021

TU Munich

Munich, Germany

- Taught 30+ lab sessions to 12+ students on real-time, multi-threaded FreeRTOS development in C.

Publications

Network Support For Scalable And High Performance Cloud Exchanges | *ACM SIGCOMM '25*

Sept 2025

- Onyx**: cloud financial exchange delivering $\leq 250 \mu\text{s}$ multicast latency, within $< 1 \mu\text{s}$ across 1K receivers.
- Kernel-bypass** datapath (**DPDK**) sustaining **175K multicast msgs/sec** at ~**50% lower latency** than AWS multicast.
- Cited by researchers and engineers from Jane Street, Google and Microsoft Research.

Projects

Cloud-TreeBuilder (*M.Sc. Thesis*) | *GCP, ZMQ, Terraform, Python, C++, Distributed Systems*

Mar 2024 – Oct 2024

- Mitigates **straggler VMs** by probing cluster network performance to build a multicast tree of depth D, fan-out F. Ranks candidate VMs by weighted **p90 one-way delay** and standard deviation from UDP probes
- Custom heuristic matches **LemonDrop's NP-hard QAP** optimizer on a 25-VM cluster: **7.7% lower p90** and **10.6% lower median** worst-receiver latency at a fraction of the compute.

Technical Skills

Languages: Rust, C++, C, Python, Golang, Bash, VHDL

Cloud: AWS (CDK, Lambda, Step Functions, DynamoDB, KMS, EventBridge), GCP, Terraform, Packer, Docker

Tools: Unix Shell, Git, GitLab CI, CMake, GNU Make, Vim, Tmux

Technologies: Linux, TCP/UDP, HSM, gRPC, Protobufs, ZeroMQ, Tokio, MPI, OpenMP, DPDK, FreeRTOS, FPGA

Education

Technical University of Munich

M.Sc. Electrical and Computer Engineering

Oct 2020 – Oct 2024

Munich, Germany

- M.Sc. Thesis: VM Selection Heuristic for Financial Exchanges in the Cloud
- Visiting Graduate Student: **New York University** (2023 – 2024) – Operating Systems Coursework, Research

Technical University of Munich

B.Sc. Electrical and Computer Engineering

Oct 2016 – Sept 2020

Munich, Germany